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## Education

### University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

**Relevant Coursework:** Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

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## Skills

**Languages:** Python, TypeScript, C++, SQL, Java

**Software:** PyTorch, React, Next.js, Node.js, AWS, SageMaker, Fargate, Docker, CI/CD, REST, Linux

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## Experience

### L3Harris Technologies

Associate Software Engineer

Greenville, TX

June 2025 – Present

- Built containerized microservices for high-frequency data ingestion, processing, and storage using Podman and serverless architecture on defense sensor streams.
- Developed signal processing algorithms in Python and C++ on embedded systems, meeting strict latency and resource constraints in collaboration with hardware teams.
- Automated CI/CD pipelines with static code analysis and integration testing, reducing deployment friction across engineering teams.

### University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

### Doxy.me

Software Engineering Intern

Charleston, SC

August 2024 – May 2025

- Built and fine-tuned transformer-based models with PyTorch and Llama for real-time clinical inference, integrating WebRTC video streaming into the inference loop.
  - Designed and deployed end-to-end ML inference pipelines on SageMaker and Fargate covering data ingestion, preprocessing, model serving, and post-processing; improved diagnostic accuracy using reinforcement learning.
  - Implemented AI-powered document processing workflows to extract and structure clinical data from unstructured inputs, enabling downstream analytics.
  - Owned CI/CD configuration across cloud infrastructure and application deployments using YAML-based pipelines, enabling reproducible environments and zero-downtime releases.
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## Projects

**University Course Planning/Review Platform (Class Best):** Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

**Telehealth Mental Health Diagnosis Tool (Doxy.me):** Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

**Configuration Fuzzer:** Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.